

EXPERIMENT 5

Aim

Socket programming for transport layer packet (UDP client)

Theory

The **UDP Client** operates as an active sender in a connectionless architecture. Because UDP does not maintain a persistent connection, the client can immediately transmit datagrams to a target server once the destination IP address and port number are known.

UDP Client Execution Steps:

1. **Socket Creation (socket()):** The client creates a datagram socket using AF_INET (IPv4) and SOCK_DGRAM (UDP).
2. **Addressing Setup:** The client configures a target address structure (sockaddr_in) containing the remote server's IP address and port number.
3. **Data Transmission (sendto()):** The client sends data directly using sendto(). Since UDP is connectionless, no connect() call is required.
4. **Data Reception (recvfrom()):** The client optionally waits for a reply from the server using recvfrom().
5. **Closing Socket (close()):** Closes the client socket descriptor upon completion.

```
lab-03-17@lab-03-17-OptiPlex-3280-AIO:~$ vim udp_client_1290.c
lab-03-17@lab-03-17-OptiPlex-3280-AIO:~$ gcc udp_client_1290.c -o udp_client_1290
lab-03-17@lab-03-17-OptiPlex-3280-AIO:~$ ./udp_client_1290
Enter message: Hello
Server Response: Hello from UDP Server
```



```
lab-03-17@lab-03-17-OptiPlex-3280-AIO: ~
#include <stdio.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>

int main() {
    int sockfd;
    char message[100];
    char buffer[100];

    struct sockaddr_in serverAddr;
    socklen_t addr_size;

    // Create UDP socket
    sockfd = socket(AF_INET, SOCK_DGRAM, 0);

    if (sockfd < 0) {
        perror("Socket creation failed");
        return 1;
    }

    // Configure server address
    serverAddr.sin_family = AF_INET;
    serverAddr.sin_port = htons(8888);
    serverAddr.sin_addr.s_addr = inet_addr("127.0.0.1");

    printf("Enter message: ");
    fgets(message, sizeof(message), stdin);

    // Send message to server
    sendto(sockfd,
           message,
           strlen(message),
           0,
           (struct sockaddr *)&serverAddr,
           sizeof(serverAddr));

    // Receive response from server
    addr_size = sizeof(serverAddr);

    int n = recvfrom(sockfd,
                     buffer,
                     sizeof(buffer) - 1,
                     0,
                     (struct sockaddr *)&serverAddr,
                     &addr_size);

    if (n < 0) {
        perror("Receive failed");
        close(sockfd);
        return 1;
    }

    buffer[n] = '\0';
    printf("Server Response: %s\n", buffer);

    // Close socket
    close(sockfd);

    return 0;
}
```

Conclusion

The UDP Client socket was successfully constructed at the Transport Layer. Using `SOCK_DGRAM` and the `sendto()` system call, the client demonstrated connectionless communication by sending datagram packets directly to the server without needing to perform a TCP-style connection handshake.